Microsoft Azure Fundamentals (AZ-900) — Lab Index

All labs run on a **free Azure account**:

https://azure.microsoft.com/free (USD 200 credit for 30 days + always-free services)

Portal labs run at **https://portal.azure.com**; CLI labs run in **Azure Cloud Shell** (https://shell.azure.com) — nothing to install on your own machine. Labs marked **Web tool** use a free public website and need no Azure subscription at all.

The labs share the resource group `rg-az900-labs` (region **Southeast Asia**). Run each lab's **Clean up** section; the full course clean-up happens at the end of Lab 16.

Domain 1 — Describe Cloud Concepts (25–30%)

#	Lab	Where it runs
1	Tour the Azure Portal and Create Your First Resource Group	Azure Portal
2	Get Started with Azure Cloud Shell (CLI and PowerShell)	Cloud Shell
3	Explore Regions, Region Pairs and Availability Zones	Cloud Shell + **Web tools** — https://datacenters.microsoft.com/globe/explore, https://www.azurespeed.com

Domain 2 — Describe Azure Architecture and Services (35–40%)

#	Lab	Where it runs
4	Create a Windows Virtual Machine in the Portal	Azure Portal + RDP client (mstsc / Windows App)
5	Create a Linux VM with the Azure CLI	Cloud Shell
6	Create a Web App with Azure App Service	Azure Portal + Cloud Shell
7	Deploy a Container with Azure Container Instances	Cloud Shell
8	Create a Virtual Network and Test Private Connectivity	Azure Portal + Cloud Shell
9	Create a Storage Account and Work with Blob Storage	Azure Portal + Cloud Shell
10	Create an Azure SQL Database	Azure Portal (Query editor)
11	Explore Microsoft Entra ID: Users and Groups	Azure Portal — Note: use your own free-account tenant

Domain 3 — Describe Azure Management and Governance (30–35%)

#	Lab	Where it runs
12	Manage Access with Azure RBAC	Azure Portal
13	Apply Resource Tags and Resource Locks	Azure Portal + Cloud Shell
14	Enforce Standards with Azure Policy	Azure Portal
15	Estimate and Control Costs: Pricing Calculator, TCO and Budgets	Azure Portal + **Web tools** — https://azure.microsoft.com/pricing/calculator/, https://azure.microsoft.com/pricing/tco/calculator/
16	Monitor Your Environment: Advisor, Service Health and Azure Monitor	Azure Portal — ends with full course clean-up

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Two-day schedule

Day	Labs	Focus
Day 1	Labs 1–8	Cloud concepts, portal & CLI tooling, compute (VMs, App Service, containers), networking
Day 2	Labs 9–16	Storage, databases, identity (Entra ID), RBAC, governance (tags, locks, policy), cost management, monitoring

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How each lab folder is laid out

Every lab is a self-contained folder:

```
lab-05-linux-vm-cli/
|--- README.md <- the lab instructions (Markdown)
|--- lab-05-linux-vm-cli.pdf <- the same instructions (PDF, for printing/handout)
|--- deploy-linux-vm.sh <- optional helper script
`--- cleanup-linux-vm.sh
```

- `README.md` — the lab instructions; this is what you follow on screen.
- `<lab-name>.pdf` — the identical instructions as a PDF handout.
- **Scripts, YAML, JSON and SQL** — optional helpers that automate the commands in the lab. They are a convenience, not a shortcut: the steps in the README teach the concepts command by command, and the AZ-900 exam tests understanding, not typing.

To use one, upload it to Cloud Shell (**Manage files > Upload**) or clone this repo there, then run `bash <script-name>`.

Regenerate every PDF after editing any lab:

```
python3 qa/render_lab_pdfs.py
```